

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter 2024

Course: B.TECH

Semester : III

Branch : Electronics & Telecommunication Engineering /Electronics Engineering

Subject Code & Name: BTETC303/BTEXC303 - Digital Electronics

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)			12
1	Which number system is used in digital electronics?	Apply	1
	Decimal Binary Octal Hexadecimal		
2	What is the decimal equivalent of the binary number 101?	Understand	1
	5 7 4 3		
3	Which numbering system uses digits and letters (A-F)?	Understand	1
	Decimal Binary Octal Hexadecimal		
4	Which gate is called a universal gate?	Analyze	1
	AND XOR NAND NOT		
5	How many bits are there in a nibble?	Understand	1
	2 4 6 8		
6	Which of the following is a combinational circuit?	Analyze	1
	FLIPFLOP Multiplexer COUNTER FSM		
7	Which circuit converts binary data to decimal?	Understand	1
	Encoder Decoder Multiplexer Demultiplexer		
8	What does a truth table represent?	Apply	1
	Circuit design Logical operations Voltage levels Input-output relationship		
9	In a JK flip-flop, what happens when both J and K inputs are HIGH?	Understand	1

	Set	Reset	Toggle	No change		
10	What is the basic building block of a digital circuit?				Apply	1
	Logic Gate	Resistor	Resistor	Encoder		
11	How many select lines are needed for a 16-to-1 multiplexer?					1
	3	4	5	6		
12	What is the minimum number of NAND gates required to implement an XOR gate?				Apply	1
	3	4	5	6		

Q.2 Solve the following.

- A) What is universal gates? Explain how NAND and NOR gates can be used to implement basic logic gates. Analyze 6
- B) Prove the following identities using Boolean algebra:
 i) $A(A+B) = A$ Apply 6
 ii) $A+AB'+AB=A+B$

Q.3 Solve the following.

- A) Minimize the following Boolean function using K-map and draw the logic circuit:
 $F(A,B,C,D) = A'B'C'D + AB'C'D' + ABCD' + AB'CD + ABC'D$ Analyze 6
- B) What is Full adder? Design Full adder using half adder. Apply 6

Q.4 Solve Any Two of the following.

- A) What is a multiplexer? Explain the working of a 4-to-1 multiplexer. Analyze 6
- B) Convert S-R flip-flop to J-K flip-flop. Understand 6
- C) Explain characteristics CMOS logical family. How CMOS can be realized as inverter? Apply 6

Q.5 Solve Any Two of the following.

- A) Design 3-bit Synchronous Down counter using D Flip Flop. Understand 6
- B) Write a note on registers. Explain in brief the classification of registers with diagrams Apply 6

- C) Implement a 2-bit magnitude comparator using basic gates.
Provide the truth table and circuit diagram Analyze 6
- Q. 6 Solve Any Two of the following.** 12
- A) Draw and explain the block diagram of BCD to 7 segment display and write the truth table. Also derive the expression for any two outputs using k-map . Analyze 6
- B) Construct a state table for a sequence detector that detects the binary sequence '101'. Apply 6
- C) Define following terms for digital ICs:
- i) Speed of operation/ Propagation delay
 - ii) Power of dissipation
 - iii) Figure of merit Understand 6
 - iv) Fan in
 - v) Fan out
 - vi) Noise immunity

*** End ***